

HydroSense

Real-Time Water Leakage Detection & Quality Monitoring System



Detects Leaks



Prevents Theft



Monitors Water Quality



Automated Response



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Background and Existing Real-World System

Existing Gap

40%

India loses an estimated **40%** of **treated water** to leakage and theft (Non-Revenue Water) in municipal pipeline networks.



Current detection is **reactive** — leaks found only via visible pooling or delayed billing anomalies.



Illegal tapping/theft goes undetected without manual physical inspection.



Water quality tested only **periodically**, not continuously — contamination often discovered only after consumption.



Current Process is Slow and Reactive



Leak Occurs
(Undetected)



Visible Pooling
or Complaint



Manual
Inspection



Repair
Action



Delayed
Billing Anomaly

Core Problem Definition

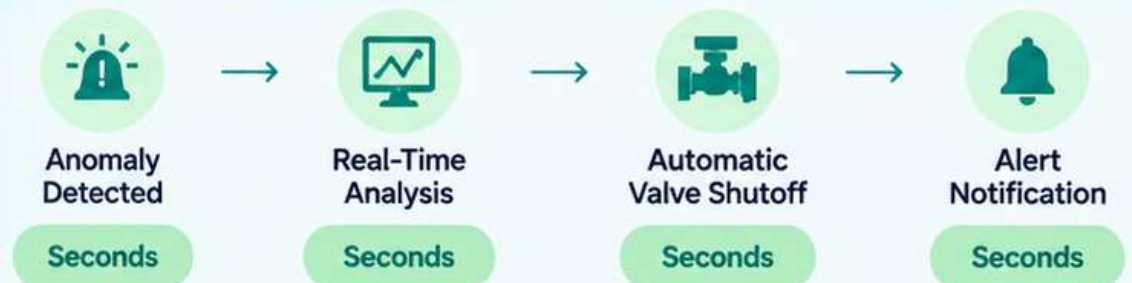


Water distribution networks lack a real-time, automated system that simultaneously detects **leakage, tampering/theft, and contamination** — and takes **corrective action** without waiting for manual inspection or complaints.

Current System – Manual and Reactive



Proposed System – Real-Time and Automated



End / Target Audience

Who Will Use HydroSense?

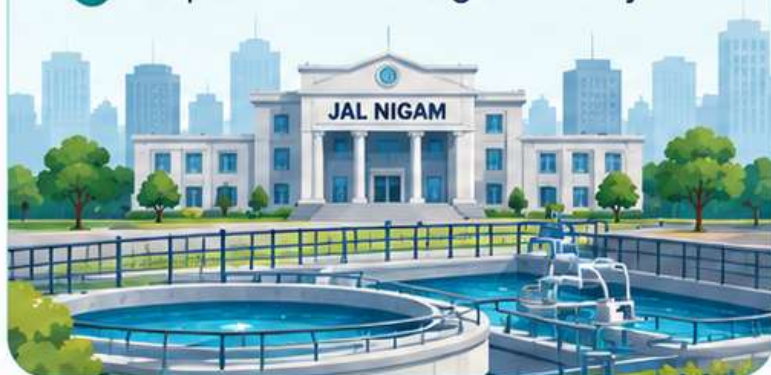
HydroSense



Primary Audience

**Municipal water boards
/ Jal Nigam departments**

- ✓ Manage city water supply
- ✓ Large pipeline networks
- ✓ Reduce non-revenue water
- ✓ Improve monitoring efficiency



Secondary Audience

**Housing societies,
campuses, industrial estates**

- ✓ Monitor internal distribution
- ✓ Detect leaks and tampering
- ✓ Ensure safe water quality
- ✓ Reduce maintenance costs



Tertiary Audience

**Rural / semi-urban
water schemes**

- ✓ Monitor rural water supply
- ✓ Detect leaks and theft
- ✓ Ensure safe drinking water
- ✓ Improve access and reliability



Scope and Limitations

What Our Project Covers and What It Does Not



In Scope



Leak detection

Using F1/F2 flow differential and drift (CUSUM)



Tamper detection

Using vibration sensor and Z-score



Water quality monitoring

pH, TDS, turbidity, and temperature



Automated valve shutoff

Relays open and solenoid valves close on confirmed leak/tamper



Limitations



Uses synthetic data

Currently in simulation stage, not real field data.



Possible false positives

Nearby construction or traffic vibration may trigger alerts.



Does not predict degradation

No long-term pipe health or failure prediction.



Needs stable power and connectivity

Each node requires reliable power and network (Wi-Fi/GSM/LoRa).

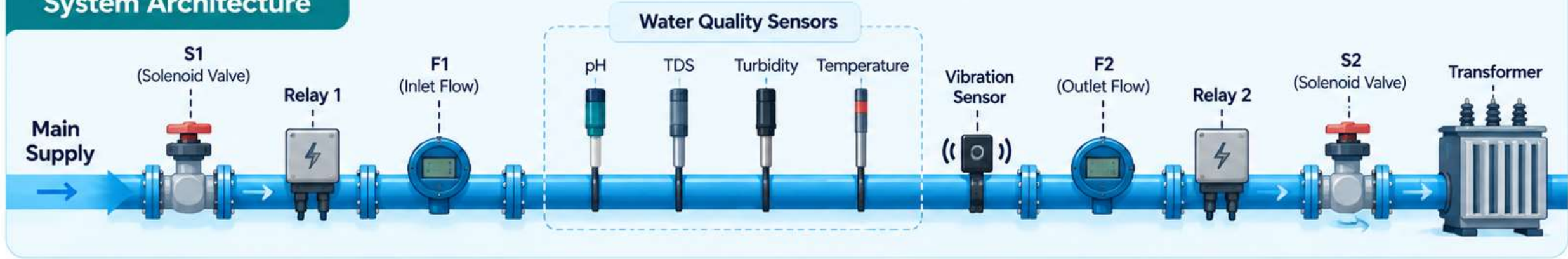


Proposed Model and Features

System Architecture, Detection Logic and Automated Response



System Architecture



Detection Logic



CUSUM

Detects leak using flow differential drift (F1 vs F2)



Isolation Forest

Anomaly detection as cross-check



Z-score

Detects vibration/tamper spikes



Automated Response



Leak or
tamper
confirmed



Relays open



Solenoid
valves shut



Supply
cut off



Prevents water loss and unauthorized usage automatically.



Key Features



Real-time leak detection
Using F1/F2 flow differential



Tamper detection
Using vibration sensor



Continuous water quality monitoring
pH, TDS, turbidity, temperature



Automated valve shutoff
On confirmed leak or tamper

Expected Outcomes and Impact

Creating Safer, Smarter and More Sustainable Water Systems

HydroSense



Fast Detection

- ✓ Detect leaks/tamper within seconds
- ✓ Vs. days/weeks manually
- ✓ Quick response to minimize loss



Reduced Water Wastage

- ✓ Less non-revenue water (NRW)
- ✓ Direct cost savings for municipalities
- ✓ Improved supply efficiency



Early Contamination Alerts

- ✓ Continuous monitoring of pH, TDS, turbidity
- ✓ Detect unsafe water before consumption
- ✓ Reduced risk of waterborne diseases



Reduced Water Theft

- ✓ Immediate tamper detection
- ✓ Prevents illegal tapping
- ✓ Ensures fair distribution and accountability



Supports Sustainable Development

- ✓ Conserves water resources
- ✓ Improves public health
- ✓ Builds resilient and sustainable cities

6 CLEAN WATER AND SANITATION



Feasibility Analysis

Practical, Cost-Effective and Scalable Solution

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Technical Feasibility



Off-the-shelf hardware

ESP32, standard sensors



Proven algorithms

CUSUM, Isolation Forest, Z-score



Open-source software stack

Python, Firebase, React



Technically implementable

Using existing, reliable components



Overall Feasibility

HIGH



Cost Feasibility



~₹2,000 – 4,000

Per sensor node (estimated)



Uses low-cost, widely available components

ESP32, sensors, relay, valve



Open-source software

No licensing costs



Low maintenance cost

Standard, replaceable sensors



Overall Feasibility

LOW COST



Deployment Feasibility



Needs stable power supply

Solvable via solar in remote areas



Needs network connectivity

Wi-Fi / GSM / LoRa options



Scalable to multiple locations

Municipal, rural and industrial areas



Can be deployed in phases

Start with high-risk zones



Overall Feasibility

HIGH

